

DAKAR, Senegal

WMO: 616410

Lat: 14.740N

Lon: 17.490W

Elev: 85

StdP: 14.65

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	62.4	63.0	33.7	28.4	77.9	41.0	37.9	77.0	23.0	70.0	21.2	70.7	10.5	0	0.418

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	10.0	91.0	72.0	88.8	76.4	87.8	77.8	82.4	85.8	81.5	85.1	80.8	84.7	8.6	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.3	163.1	84.4	80.7	159.8	83.9	79.7	154.6	83.4	46.5	85.5	45.3	85.6	44.5	84.7	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.7	18.8	17.3	DB	60.0	98.3	2.1	2.6	58.5	100.2	57.2	101.7	56.0	103.2	54.5	105.1
			WB	49.4	84.3	2.7	1.6	47.4	85.4	45.8	86.4	44.3	87.3	42.3	88.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	77.1	72.3	71.1	71.5	71.8	74.3	79.0	81.5	81.9	82.2	82.8	80.3	76.2
	DBStd	5.25	3.79	3.88	3.36	2.61	2.56	2.31	1.69	1.69	1.81	1.99	2.59	3.24
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	9895	690	592	665	655	752	869	978	989	966	1017	909	813
	CDD65	4421	225	172	201	205	287	419	513	524	516	552	459	348
	CDH74	35446	743	569	605	462	1028	3216	5293	5718	5723	6239	4083	1767
	CDH80	9545	165	153	136	57	91	582	1333	1600	1779	2153	1123	373
(14) Wind	WSAvg	9.5	10.9	11.7	12.2	12.4	10.7	8.2	7.5	7.1	6.3	7.2	9.5	10.4
(15) Precipitation	PrecAvg	16.7	0.0	0.0	0.0	0.0	0.0	0.3	2.3	6.7	5.4	1.2	0.0	0.0
	PrecMax	35.6	0.5	1.6	0.4	0.0	0.0	3.0	8.7	16.3	14.6	7.2	0.8	0.7
	PrecMin	4.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.8	1.6	0.0	0.0	0.0
	PrecStd	6.8	0.1	0.2	0.0	0.0	0.0	0.6	2.0	3.6	2.9	1.7	0.1	0.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.9	91.6	91.2	86.0	84.6	87.9	88.3	88.2	89.9	92.1	94.6	91.5
		MCWB	61.6	64.5	62.8	68.9	73.9	77.4	78.4	79.4	80.2	77.3	68.5	64.7
	2%	DB	84.3	84.3	82.8	80.5	82.4	85.9	87.4	87.5	88.1	89.7	89.9	86.4
		MCWB	62.9	63.5	65.7	70.1	73.3	77.0	78.2	79.5	80.4	79.1	72.2	67.2
	5%	DB	80.5	80.2	79.3	78.3	80.5	84.4	86.1	86.3	87.5	88.2	87.5	84.0
		MCWB	64.0	64.1	66.7	69.8	72.9	76.5	77.9	79.2	80.4	79.6	73.8	67.7
	10%	DB	77.4	76.9	76.9	76.6	78.8	83.0	85.2	85.8	86.2	87.6	85.7	81.2
		MCWB	64.4	64.2	66.8	69.5	72.2	75.8	77.5	79.1	80.0	79.6	73.9	68.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.7	71.7	72.4	73.7	77.4	80.1	81.0	83.0	83.8	83.7	80.2	77.0
		MCDB	76.8	77.3	77.8	78.4	81.6	83.9	84.7	85.4	86.7	87.5	84.6	80.6
	2%	WB	70.2	69.7	70.8	71.9	75.5	78.9	80.0	81.5	82.3	82.5	79.0	75.1
		MCDB	74.8	75.0	76.0	76.9	79.6	82.9	84.2	84.5	85.6	86.4	83.7	78.9
	5%	WB	69.0	68.6	69.6	71.0	74.0	77.9	79.2	80.6	81.4	81.4	78.0	73.8
		MCDB	73.9	73.5	74.7	75.8	78.3	82.2	83.6	84.2	85.1	85.5	82.8	78.0
	10%	WB	67.7	67.4	68.7	70.1	72.8	76.9	78.6	79.9	80.7	80.7	77.0	72.5
		MCDB	73.1	72.1	73.6	74.7	77.2	81.6	83.2	83.6	84.6	84.9	81.7	77.1
(35) Mean Daily Temperature Range	5% DB	MDBR	12.1	12.3	11.6	10.0	9.0	8.5	8.0	8.0	9.2	10.0	11.2	11.4
		MCDBR	17.0	19.0	16.6	12.7	10.6	9.4	8.9	8.7	10.0	11.1	15.8	16.1
		MCWBR	10.6	7.9	6.5	5.6	5.0	4.7	4.8	5.1	5.8	6.0	9.0	11.0
	5% WB	MCDBR	12.6	12.0	11.8	10.8	9.5	8.9	8.5	8.4	9.4	10.2	12.0	11.9
		MCWBR	8.1	6.3	6.1	5.9	5.5	5.6	5.4	5.7	6.2	6.2	7.2	7.7
(40) Clear-Sky Solar Irradiance	taub		0.529	0.551	0.616	0.630	0.687	0.739	0.692	0.621	0.595	0.601	0.523	0.509
	taud		1.911	1.869	1.743	1.769	1.661	1.640	1.746	1.912	1.957	1.907	2.025	2.012
	Ebn at Noon		242	244	232	228	213	199	210	227	234	228	243	244
	Edh at Noon		60	65	75	73	80	81	73	62	60	62	53	53
(44) All-Sky Solar Radiation	RadAvg		1659	1848	2154	2268	2242	2093	1874	1797	1762	1824	1656	1561
	RadStd		109	113	75	65	82	49	76	85	77	93	98	74

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page